PRATIKUM PBO (Pemrograma Beroruentasi Objek)

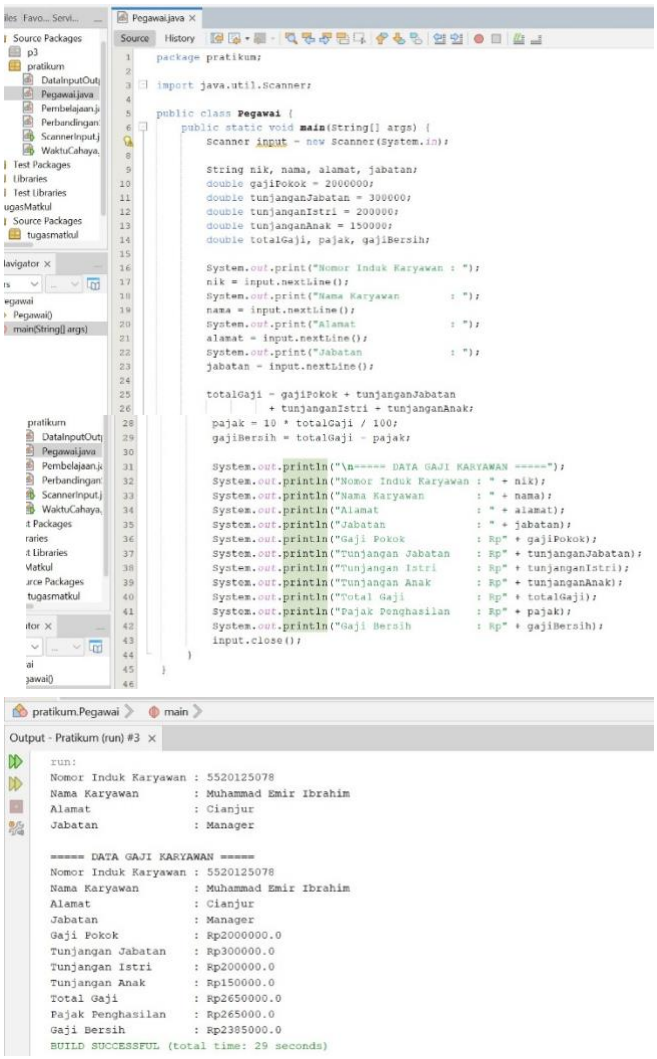
Nama : Muhammad Emir Ibrahim

Kelas : IF C 25

Npm : 5520125078

Shift : 5

1. Gaji Bersih Seorang Pegawai



```
package pratikum;

import java.util.Scanner;

public class Pegawai {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        String nik, nama, alamat, jabatan;
        double gajiPokok = 2000000;
        double tunjanganJabatan = 300000;
        double tunjanganIstri = 200000;
        double tunjanganAnak = 150000;
        double totalGaji, pajak, gajiBersih;

        System.out.print("Nomor Induk Karyawan : ");
        nik = input.nextLine();
        System.out.print("Nama Karyawan       : ");
        nama = input.nextLine();
        System.out.print("Alamat              : ");
        alamat = input.nextLine();
        System.out.print("Jabatan             : ");
        jabatan = input.nextLine();

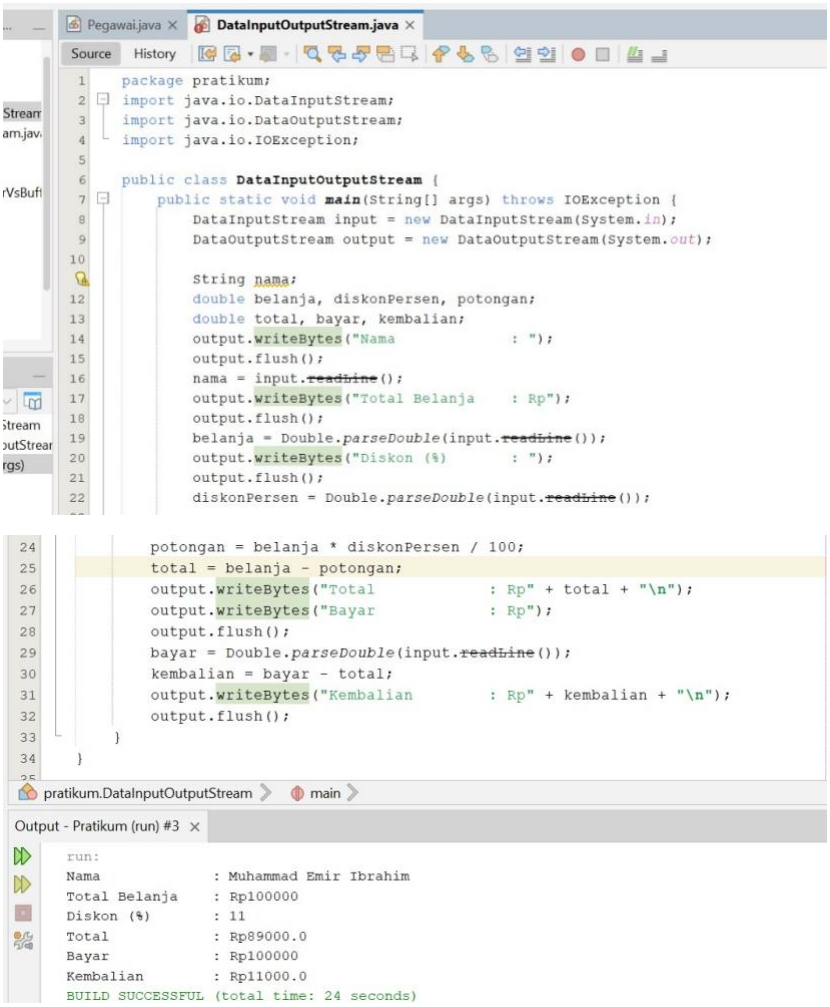
        totalGaji = gajiPokok + tunjanganJabatan
                  + tunjanganIstri + tunjanganAnak;
        pajak = 10 * totalGaji / 100;
        gajiBersih = totalGaji - pajak;

        System.out.println("\n===== DATA GAJI KARYAWAN =====");
        System.out.println("Nomor Induk Karyawan : " + nik);
        System.out.println("Nama Karyawan        : " + nama);
        System.out.println("Alamat               : " + alamat);
        System.out.println("Jabatan              : " + jabatan);
        System.out.println("Gaji Pokok           : Rp" + gajiPokok);
        System.out.println("Tunjangan Jabatan    : Rp" + tunjanganJabatan);
        System.out.println("Tunjangan Istri      : Rp" + tunjanganIstri);
        System.out.println("Tunjangan Anak       : Rp" + tunjanganAnak);
        System.out.println("Total Gaji           : Rp" + totalGaji);
        System.out.println("Pajak Penghasilan    : Rp" + pajak);
        System.out.println("Gaji Bersih          : Rp" + gajiBersih);
        input.close();
    }
}
```

```
run:
Nomor Induk Karyawan : 5520125078
Nama Karyawan        : Muhammad Emir Ibrahim
Alamat               : Cianjur
Jabatan              : Manager

===== DATA GAJI KARYAWAN =====
Nomor Induk Karyawan : 5520125078
Nama Karyawan        : Muhammad Emir Ibrahim
Alamat               : Cianjur
Jabatan              : Manager
Gaji Pokok           : Rp2000000.0
Tunjangan Jabatan    : Rp300000.0
Tunjangan Istri      : Rp200000.0
Tunjangan Anak       : Rp150000.0
Total Gaji           : Rp2650000.0
Pajak Penghasilan    : Rp265000.0
Gaji Bersih          : Rp2385000.0
BUILD SUCCESSFUL (total time: 29 seconds)
```

2. Program Java Untuk DataInputStream Dan DataOutputStream



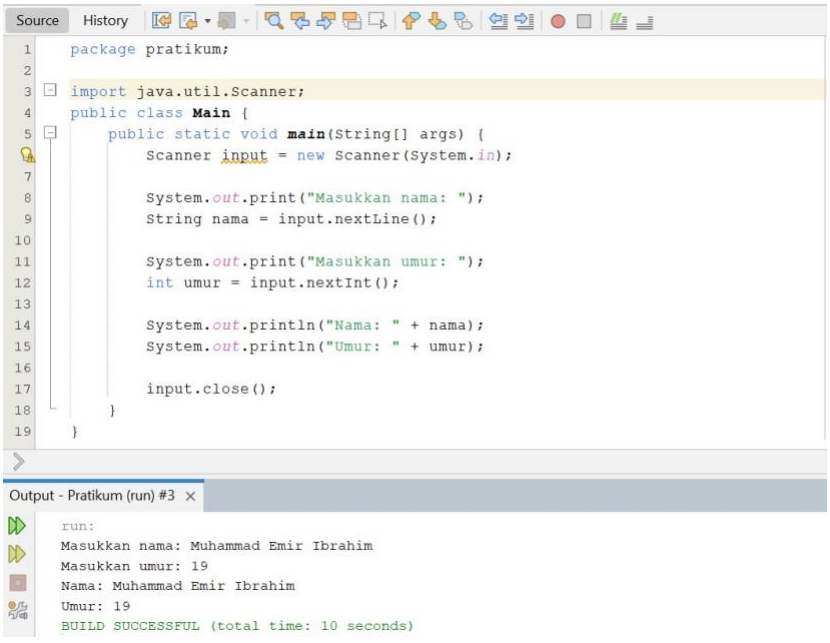
```
package pratikum;
import java.io.DataInputStream;
import java.io.DataOutputStream;
import java.io.IOException;

public class DataInputOutputStream {
    public static void main(String[] args) throws IOException {
        DataInputStream input = new DataInputStream(System.in);
        DataOutputStream output = new DataOutputStream(System.out);

        String nama;
        double belanja, diskonPersen, potongan;
        double total, bayar, kembalian;
        output.writeBytes("Nama             : ");
        output.flush();
        nama = input.readline();
        output.writeBytes("Total Belanja    : Rp");
        output.flush();
        belanja = Double.parseDouble(input.readline());
        output.writeBytes("Diskon (%)       : ");
        output.flush();
        diskonPersen = Double.parseDouble(input.readline());
        potongan = belanja * diskonPersen / 100;
        total = belanja - potongan;
        output.writeBytes("Total            : Rp" + total + "\n");
        output.writeBytes("Bayar            : Rp");
        output.flush();
        bayar = Double.parseDouble(input.readLine());
        kembalian = bayar - total;
        output.writeBytes("Kembalian        : Rp" + kembalian + "\n");
        output.flush();
    }
}
```

```
run:
Nama             : Muhammad Emir Ibrahim
Total Belanja    : Rp100000
Diskon (%)       : 11
Total            : Rp89000.0
Bayar            : Rp100000
Kembalian        : Rp11000.0
BUILD SUCCESSFUL (total time: 24 seconds)
```

3. Class Scanner Dan Class BufferReader



```
package pratikum;

import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Masukkan nama: ");
        String nama = input.nextLine();

        System.out.print("Masukkan umur: ");
        int umur = input.nextInt();

        System.out.println("Nama: " + nama);
        System.out.println("Umur: " + umur);

        input.close();
    }
}
```

```
run:
Masukkan nama: Muhammad Emir Ibrahim
Masukkan umur: 19
Nama: Muhammad Emir Ibrahim
Umur: 19
BUILD SUCCESSFUL (total time: 10 seconds)
```

```java
package pratikum;

import java.io.*;
public class Main {
    public static void main(String[] args) throws IOException {
        BufferedReader input = new BufferedReader(new InputStreamReader(System.i

        System.out.print("Masukkan nama: ");
        String nama = input.readLine();

        System.out.print("Masukkan umur: ");
        int umur = Integer.parseInt(input.readLine());

        System.out.println("Nama: " + nama);
        System.out.println("Umur: " + umur);
    }
}
```

```
Output - Pratikum (run) #3
run:
Masukkan nama: Muhammad Emir Ibrahim
Masukkan umur: 19
Nama: Muhammad Emir Ibrahim
Umur: 19
BUILD SUCCESSFUL (total time: 11 seconds)
```

4. Program Menghitung Gaya Sentripetal

```java
package pratikum;

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        double M, v, r, Fc;

        System.out.print("Masukkan massa (kg): ");
        M = input.nextDouble();
        System.out.print("Masukkan kecepatan (m/s): ");
        v = input.nextDouble();
        System.out.print("Masukkan jari-jari (m): ");
        r = input.nextDouble();
        Fc = (M * v * v) / r;
        System.out.println("Gaya sentripetal = " + Fc + " Newton");
        input.close();
    }
}
```

```
Output - Pratikum (run) #3
run:
Masukkan massa (kg): 100
Masukkan kecepatan (m/s): 900
Masukkan jari-jari (m): 2
Gaya sentripetal = 4.05E7 Newton
BUILD SUCCESSFUL (total time: 11 seconds)
```

5. Menggunakan Program Java, Menggunakan BufferReadber

```java
package pratikum;
import java.io.*;

public class Main {
    public static void main(String[] args) throws IOException {
        BufferedReader input = new BufferedReader(
            new InputStreamReader(System.in)
        );

        int mangga, jambu, manggaBusuk;
        int totalMangga, totalJambu, totalBuah;
        System.out.print("Jumlah mangga yang dibeli: ");
        mangga = Integer.parseInt(input.readLine());
        System.out.print("Jumlah jambu yang dibeli: ");
        jambu = Integer.parseInt(input.readLine());
        System.out.print("Jumlah mangga busuk: ");
        manggaBusuk = Integer.parseInt(input.readLine());

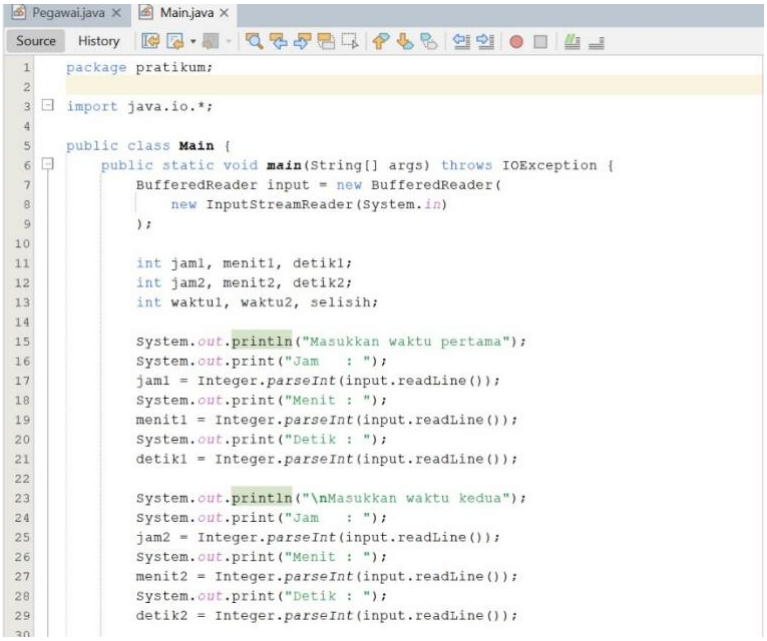
        totalMangga = mangga - manggaBusuk;
        totalJambu = jambu;
        totalBuah = totalMangga + totalJambu;

        System.out.println("\n===== HASIL PEMBELIAN =====");
        System.out.println("Mangga yang dibeli     : " + mangga + " pcs");
        System.out.println("Mangga yang busuk      : " + manggaBusuk + " pcs");
        System.out.println("Mangga yang diterima   : " + totalMangga + " pcs");
        System.out.println("Jambu yang dibeli      : " + jambu + " pcs");
        System.out.println("Total buah diterima    : " + totalBuah + " pcs");
    }
}
```

```
Output - Pratikum (run) #3
run:
Jumlah mangga yang dibeli: 100
Jumlah jambu yang dibeli: 1000
Jumlah mangga busuk: 25

===== HASIL PEMBELIAN =====
Mangga yang dibeli     : 100 pcs
Mangga yang busuk      : 25 pcs
Mangga yang diterima   : 75 pcs
Jambu yang dibeli      : 1000 pcs
Total buah diterima    : 1075 pcs
BUILD SUCCESSFUL (total time: 17 seconds)
```

6. Program Java



```
package pratikum;

import java.io.*;

public class Main {
    public static void main(String[] args) throws IOException {
        BufferedReader input = new BufferedReader(
            new InputStreamReader(System.in)
        );

        int jam1, menit1, detik1;
        int jam2, menit2, detik2;
        int waktu1, waktu2, selisih;

        System.out.println("Masukkan waktu pertama");
        System.out.print("Jam   : ");
        jam1 = Integer.parseInt(input.readLine());
        System.out.print("Menit : ");
        menit1 = Integer.parseInt(input.readLine());
        System.out.print("Detik : ");
        detik1 = Integer.parseInt(input.readLine());

        System.out.println("\nMasukkan waktu kedua");
        System.out.print("Jam   : ");
        jam2 = Integer.parseInt(input.readLine());
        System.out.print("Menit : ");
        menit2 = Integer.parseInt(input.readLine());
        System.out.print("Detik : ");
        detik2 = Integer.parseInt(input.readLine());

        waktu1 = jam1 * 3600 + menit1 * 60 + detik1;
        waktu2 = jam2 * 3600 + menit2 * 60 + detik2;
        selisih = Math.abs(waktu2 - waktu1);

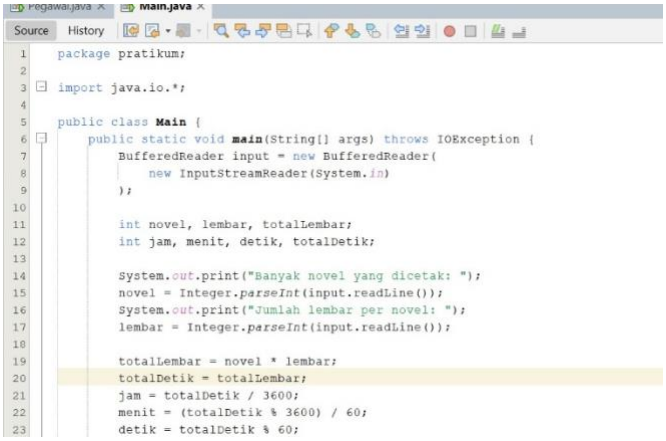
        System.out.println("\nSelisih waktu:");
        System.out.println("Jam   : " + selisih / 3600);
        System.out.println("Menit : " + (selisih % 3600) / 60);
        System.out.println("Detik : " + selisih % 60);
    }
}
```

```
run:
Masukkan waktu pertama
Jam   : 10
Menit : 59
Detik : 50

Masukkan waktu kedua
Jam   : 17
Menit : 100
Detik : 11100

Selisih waktu:
Jam   : 10
Menit : 45
Detik : 10
BUILD SUCCESSFUL (total time: 1 minute 0 seconds)
```

7. Cetak Novel, Cetak Mesin



```
package pratikum;

import java.io.*;

public class Main {
    public static void main(String[] args) throws IOException {
        BufferedReader input = new BufferedReader(
            new InputStreamReader(System.in)
        );

        int novel, lembar, totalLembar;
        int jam, menit, detik, totalDetik;

        System.out.print("Banyak novel yang dicetak: ");
        novel = Integer.parseInt(input.readLine());
        System.out.print("Jumlah lembar per novel: ");
        lembar = Integer.parseInt(input.readLine());

        totalLembar = novel * lembar;
        totalDetik = totalLembar;
        jam = totalDetik / 3600;
        menit = (totalDetik % 3600) / 60;
        detik = totalDetik % 60;

        System.out.println("\n===== WAKTU PERCETAKAN =====");
        System.out.println("Banyak novel       : " + novel);
        System.out.println("Lembar per novel   : " + lembar);
        System.out.println("Total lembar       : " + totalLembar);
        System.out.println("Waktu yang diperlukan:");
        System.out.println("Jam                : " + jam);
        System.out.println("Menit              : " + menit);
        System.out.println("Detik              : " + detik);
    }
}
```

```
run:
Banyak novel yang dicetak: 10
Jumlah lembar per novel: 100

===== WAKTU PERCETAKAN =====
Banyak novel       : 10
Lembar per novel   : 100
Total lembar       : 1000
Waktu yang diperlukan:
Jam                : 0
Menit              : 16
Detik              : 40
BUILD SUCCESSFUL (total time: 8 seconds)
```